Sprout 2.0: Product Design & Strategy Case Study

Designing an Accessible, Trust-First Investing Journey for New Investors in India

Author: Senior Product & Front-End Designer

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1. TL;DR: The Three Biggest Decisions

1. **Reconciled the Broken KYC Funnel Beyond the Form Itself**

The brief reported a 62% drop-off at the KYC step, but reconciling the absolute numbers revealed that **~41% of downloaders (roughly 16,300 people) never reached KYC at all**, while another **14,700 abandoned mid-verification**. Instead of treating KYC as an isolated form problem, I redesigned the pre-KYC experience on both website and app: introducing an upfront "Keep these ready" preparation screen (resolving Customer Support's top complaint) and setting predictable time expectations before users touch sensitive documents.

2. **Resolved the Executive-Marketing Impasse with a Restrained Mascot Architecture**

Marketing demanded a cartoon mascot, video, live ticker, and three pop-ups to drive viral engagement; the CEO insisted on an Apple-like minimal interface with zero cartoons to protect investor trust. I broke the deadlock by designing **"Ankur" (Hindi for sprout)**: a geometric, flat seedling that acts as a quiet, respectful companion at moments of emotional friction (such as verification wait states), but is strictly banned from appearing near legal risk disclaimers, high-stress errors, or promotional pop-ups. Marketing's live ticker was cut entirely to prevent market anxiety, pop-ups were eliminated, and the video was converted into a lightweight, click-to-load poster.

3. **Engineered Hard Compliance and Low-End Performance Baselines**

With 91% of users on mobile and 40% on smartphones under ₹10,000 facing 6.2-second average 4G loads, heavy JavaScript frameworks were rejected. The entire prototype is vanilla HTML/CSS/JS with zero external CDN dependencies, self-hosted system fonts with native Devanagari support, and sub-3 KB inline SVGs. Measured homepage payload is **86.6 KB** (well under the 400 KB target and 1.5 MB ceiling). Every screen featuring investment content presents a regulatory risk disclaimer rendered at **13px** (exceeding the 12px legal minimum) visible in the first viewport without scrolling.

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2. Problem & Funnel: Mathematical Reconciliation

The Reported Data

- Total App Downloads: **40,000**

- Completed KYC: **9,000** (22.5% of total downloads)

- Made First Investment: **3,000** (7.5% of total downloads; 33.3% of KYC-completed users)

- Post-KYC Leakage: **66.7%** of verified users do not invest.

- Reported KYC Drop-off: **62%**

Reconciling the Numbers (Inference)

The brief states a 62% drop-off at the KYC step. If 62% of users who *reached* KYC abandoned it, then the completion rate among KYC starters was:

$$\text{Completion Rate} = 100\% - 62\% = 38\% \quad (0.38)$$

If 9,000 users successfully completed KYC, the implied volume of users who reached the KYC entrypoint is:

$$\text{Users Reaching KYC} = \frac{9,000}{0.38} \approx 23,684 \quad (\text{roughly } 23,700)$$

This derivation reveals two distinct failure points:

1. **At-the-Step Abandonment:** About $23,700 - 9,000 = 14,700$ users initiated KYC but abandoned before completion.

2. **Pre-KYC Leakage:** About $40,000 - 23,700 = 16,300$ downloaders (**40.8% ≈ 41%**) abandoned the app before ever encountering the KYC step.

> **NOTE:** > **Analytical Note:** This calculation assumes that the 62% drop-off rate and the 40,000 / 9,000 cohort figures represent the same measurement period. I have documented this as an inference. Before production release, product analytics must confirm exact step-by-step event conversions (`app_open` $\to$ `screen1_view` $\to$ `screen2_view` $\to$ `screen3_prep_view` $\to$ `kyc_start` $\to$ `kyc_complete` $\to$ `order_confirm`).

```
40,000 Total Downloads
│
├── 16,300 (40.8%) dropped before KYC entrypoint [Addressed by Screen 1 & 2 clarity]
│
└── 23,700 reached KYC step
    │
    ├── 14,700 (62.0%) abandoned during KYC [Addressed by Screen 3 prep & Screen 4 wait state]
    │
    └── 9,000 completed KYC
        │
        ├── 6,000 (66.7%) leaked without investing [Addressed by Screen 5 ₹100 default]
        │
        └── 3,000 made first investment (7.5% of downloads)
```

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3. What the Data Does and Does Not Say

1. Bounce Rate (78%) is Contaminated

The 78% homepage bounce rate was measured over a 2-week period that coincided with a paid acquisition ad campaign. Paid ad traffic frequently brings low-intent visitors with misaligned expectations. **What the data does NOT say:** It does not tell us the organic baseline bounce rate.

Action: Marketing and Analytics must segment bounce rate by traffic acquisition channel (Organic Search, Direct, Paid Social, Referrals) before treating 78% as an operational baseline.

2. KYC Drop-off Cause is Statistically Unknown, but Contextually Evidenced

The quantitative telemetry reports a 62% drop-off but fails to log drop-off reasons. However, qualitative customer support logs provide two vital clues:

- "*I didn't know I needed my PAN and Aadhaar ready.*"

- "*The app asked me to wait and I thought it crashed.*"

Hypotheses & Instrumentation Plan:

1. **Unprepared Documents:** Users hit the PAN/Aadhaar prompt, don't have cards in hand, exit the app, and forget to return. *Instrumentation:* Log time-on-screen for Screen 3 and track click rate on "Remind me later".

2. **Registry Latency Mistaken for Crash:** UIDAI/government registry verification taking 30-60 seconds causes users to force-quit. *Instrumentation:* Track exit timestamps against registry response latency.

3. **Low-End Hardware Failure:** Camera crashing during document scan or slow 4G timeouts. *Instrumentation:* Track network socket error codes and device RAM telemetry.

4. **Distrust of PAN Collection:** Reluctance to share tax IDs on an unfamiliar app. *Instrumentation:* Track click-through on the privacy and regulatory consent disclosures.

3. Device & Connectivity Realities

- **91% Mobile, 40% on Devices < ₹10,000:** Budget Android handsets (e.g. Redmi 9A, Realme C-series, Samsung M04) operate with 2GB to 3GB RAM and low single-core CPU speeds.

- **6.2s Average Load on 4G:** Heavy single-page frameworks (React/Vue with 300KB+ runtime bundles) stall main-thread parsing on budget chipsets.

Design Response: Absolute rejection of heavy JavaScript frameworks, client-side hydration, and uncompressed assets.

4. Localisation Demand is Directional ("Many" Prefer Hindi)

"Many" is not a metric. However, in Tier-2/3 India, digital financial literacy is heavily constrained by English terminology. Rather than waiting for a full localization mandate, Sprout is architected with complete string extraction (`en.json`, `hi.json`) and tested with a functional Hindi toggle on the document preparation screen.

5. Survey (n=40) Suffers from Survivorship Bias

While 65% of respondents stated "Investing feels scary", all 40 participants were *existing active users*. They represent survivors who managed to overcome the initial fear.

Action: Recommend conducting structured 30-minute interviews with **5 to 8 churned users** who dropped off at the KYC step to uncover unspoken emotional blockers.

What I Would Measure First: Success Metrics Table

Metric	Current Baseline	90-Day Target	Measurement Methodology
Organic Homepage Bounce Rate	78% (contaminated)	**< 48%**	Segmented Google Analytics 4 / PostHog by UTM medium.
Pre-KYC Drop-off Rate	~41% (16,300 users)	**< 20%**	Funnel tracking: `app_open` $\to$ `prep_complete`.
KYC Step Drop-off	62% (14,700 users)	**< 35%**	Funnel tracking: `kyc_initiated` $\to$ `kyc_approved`.
Post-KYC First Investment Leak	66.7% (6,000 users)	**< 30%**	Conversion rate from `kyc_approved` $\to$ `first_order_settled`.
Total Downloads to Investor	7.5% (3,000 users)	**> 18%**	End-to-end cohort conversion over 30 days.
Support Tickets: "App Crashed"	Baseline volume	**-70% reduction**	Customer support ticket tagging in Zendesk / Freshdesk.

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4. Target Personas [Explicitly Labelled as Research Assumptions]

> *[!IMPORTANT]*
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> ***Methodological Disclosure:*** *These personas are* ***unvalidated design assumptions*** *formulated from Tier-2/3 market realities and customer support inquiry patterns. They represent working archetypes that must be formally validated through structured qualitative interviews with 5 to 8 churned users.*



Sprout User Personas (Labelled as Research Assumptions)

Persona Archetypes Summary

1. **Rohit Verma (23, Junior ITI Technician, Bhopal) — Assumption 1:**

- *Hardware & Context:* Redmi 9A (₹7,499, 2GB RAM) on prepaid 4G. Hindi primary speaker.

- *Anxiety:* Earns ₹18,000/month; fears digital fraud and believes investing is only for wealthy metro elites.

- *Drop-off Cliff:* Unexplained loading spinners cause him to believe the app crashed or compromised his data.

- *Design Response:* Native Hindi toggle, "Keep ready" prep screen, steady wait state mascot, 86.8 KB ultra-lightweight page weight.

2. **Priya Nair (20, College Student, Kochi) — Assumption 2:**

- *Hardware & Context:* Realme C21 (₹8,999, 3GB RAM). English and Malayalam speaker.

- *Anxiety:* Has ₹100–300 saved from monthly pocket money. Intrigued by financial literacy videos but intimidated by jargon.

- *Drop-off Cliff:* Abrupt demand for PAN and Aadhaar without an explanation of purpose.

- *Design Response:* Screen 2 "I'm just curious" entry, explicit regulatory consent card, ₹100 pre-selected default.

3. **Ananya Sharma (27, Operations Associate, Jaipur) — Assumption 3:**

- *Hardware & Context:* Samsung Galaxy M04 (₹8,499, 3GB RAM). English and Hindi speaker.

- *Anxiety:* Cautious saver; fears sudden penalties or lock-ins preventing her from withdrawing emergency money.

- *Drop-off Cliff:* Inability to verify whether funds can be liquidated instantly.

- *Design Response:* Prominent "No minimum lock-in" badge, transparent portfolio breakdown, ₹100 micro-investing.

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5. Usability Test Plan (Empirical Validation Protocol)

To validate whether the Sprout 2.0 interface eradicates the evidenced customer support complaints and pre-KYC leakage, the following empirical test protocol is established:

Test Parameters

- **Cohort:** 6 participants (3 English primary, 3 Hindi primary).

- **Inclusion Criteria:** First-time investors (0 prior mutual fund folios), owning smartphones under ₹12,000, accessing via mobile 4G.

- **Format:** 45-minute moderated remote or lab session using the Think-Aloud protocol on physical Android handsets.

Core Evaluative Tasks

1. **First-Impression & Risk Comprehension (Homepage):**

- *Prompt:* "Look at this page for 15 seconds. In your own words, what is this app, how much do you need to start, and can you lose money?"

- *Target:* Immediate comprehension of ₹100 threshold and market risk notice without scrolling.

2. **Goal Selection & Expectation Setting (Screens 1 & 2):**

- *Prompt:* "Start the process. Choose what matches your intention."

- *Target:* Seamless navigation through the 7-minute expectations card with zero hesitation.

3. **Document Readiness & Fallback (Screen 3):**

- *Prompt:* "You arrive at this screen, but you realize your physical PAN card is in another room. What do you do?"

- *Target:* Spontaneous discovery and use of the "Remind me later" permission drawer without abandoning.

4. **Verification Latency & Error States (Screen 4):**

- *Prompt:* "Enter your details and press verify. The screen is now loading."

- *Target:* Zero participants force-quitting or stating "the app crashed" during the 45-second animated registry check.

5. **First Investment Settlement (Screen 5):**

- *Prompt:* "Review your starting order and complete the action."

- *Target:* Confidence in order placement without fear of hidden recurring debits.

Quantitative Success Benchmarks

- **Task Completion Rate:** $\ge 85\%$ across all 5 screens without moderator intervention.

- **Document Prep Comprehension:** $\ge 90\%$ understanding why PAN/Aadhaar are required.

- **Crash Perception Reduction: 0%** of participants believing the app crashed during registry wait states.

- **Single Ease Question (SEQ):** Average score $\ge 6.2 / 7.0$ across all completed flows.

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6. Product Design Principles

1. **Honesty Over Hype**

Never promise guaranteed returns, never display synthetic growth tickers, and never use artificial countdown timers. Financial trust is built when an app is candid about market risks.

2. **One Job Per Screen**

New investors experience cognitive overload when confronted with multi-input forms. Each screen has a single purpose, a clear progress indicator, and an unmistakable primary action.

3. **Respect for Low-End Hardware**

Design for the user on a ₹7,500 phone with 2GB RAM on an intermittent 4G connection. Every millisecond of latency is an invitation to abandon.

4. **Human Guidance Without Condescension**

Acknowledge that financial fear is normal. Never blame the user for system errors, failed document lookups, or network drops.

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7. Stakeholder Negotiation Table

Stakeholder	What Was Asked	What is Valid	My Call as Designer	Trade-off / Compromise Rationale
CEO	"Minimal, Apple-like, no cartoon characters. Investors won't trust a cartoon."	Financial apps must convey institutional credibility and stability; childlike cartoons undermine serious trust.	**The mascot stays, but as a restrained geometric seedling.** Calm, grounded vector; flat colors; no 3D, no sparkles.	Humanizes cold financial onboarding without feeling like a children's game. Prohibited from appearing near legal risk text.
Head of Marketing	Mascot, video, live ticker, 3 pop-up modals, and referral banner.	Social proof, clear product demonstration, and viral referral loops are proven acquisition levers.	**Cut the ticker and all pop-ups.** Video converted to a click-to-load poster. Referral placed as a quiet footer strip and after first investment.	Tickers induce anxiety for beginners; modals cause immediate mobile abandonment. Click-to-load video saves 4G data.
Compliance Officer	Risk disclaimer $\ge$12px visible without scrolling in first viewport. Banned terms: "guaranteed", "safe", "risk-free".	Regulatory mandates under SEBI are non-negotiable; violations risk platform suspension.	**Implemented 13px disclaimer component in the first viewport** across all investment screens. Automated copy linter runs in CI.	Disclaimer placed right below the primary CTA so users see risk transparency immediately before acting.
Lead Developer	Page weight $<1.5\text{ MB}$, CSS animations only, 1 front-end engineer for 3 weeks.	A single engineer cannot build complex state engines or debug heavy animation libraries within 21 days.	**Pure vanilla HTML/CSS/JS with zero build step.** System fonts with Devanagari fallback; total payload is **86.8 KB**.	3-week delivery plan with strict cut list protects engineering sanity and guarantees lightning performance.
Customer Support Lead	Complaints: "Didn't know I needed documents ready" and "Thought app crashed during wait".	Unprepared users drop off permanently; blank loading spinners look like software deadlocks.	**Added "Keep these ready" prep screen** and an animated 45-second wait state with reassurance text.	Directly targets and eliminates the two largest evidenced sources of user friction and support churn.

8. End-to-End User Journey Map

The journey map below traces the emotional transition of a first-time investor from initial skepticism upon seeing an ad to confident ownership of their first ₹100 portfolio.



Sprout User Journey Map

Design Responses by Journey Stage

1. **Ad Encounter:** Avoided hyperbolic claims; ad links directly to a plain-language landing page with pre-set parameters.

2. **Homepage Landing:** Replaced complex market data with an interactive 4-step vertical roadmap mirroring the app.

3. **Intent Confirmation (Screen 2):** Allowed users to declare "I'm just curious", removing pressure to make immediate financial commitments.

4. **Document Preparation (Screen 3):** Provided an explicit checklist with a "Remind me later" option that defers push notification permissions until needed.

5. **Verification Wait (Screen 4):** Displayed a calm Ankur mascot with an honest 45-second countdown and clear guidance that users can background the app without losing progress.

6. **First Investment (Screen 5):** Pre-selected ₹100 as the starting default, paired with a subtle botanical sprout animation rather than cheap gamified confetti.

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9. Homepage Design Decisions

The homepage is structured as a five-stage narrative: **Fear $\to$ Understanding $\to$ Small First Step $\to$ Trust $\to$ Action.**



Sprout Mobile Homepage Screenshot



Sprout Desktop Homepage Screenshot

1. **First Viewport Architecture (Zero-Scroll Disclaimer Compliance):**

- Compact brand header with an instant language toggle (`EN | हिंदी`).

- One headline devoid of jargon: "*Start small. Learn as you grow.*"

- Primary CTA: "*Start with ₹100*" accompanied by Ankur waving in a small 44px badge.

- Compliance Risk Disclaimer: Rendered in high-contrast text (`#3F4E46` on `#FFFFFF`, 8.78:1 ratio) at **13px**, strictly exceeding the 12px legal threshold and fully visible on both a 360px mobile viewport and desktop without scrolling (top offset 418.8px).

2. **Three Reassurance Anchors:**

- *Start with just ₹100:* Validates that small amounts matter.

- *See what you own:* Promises complete portfolio transparency.

- *Help in your language:* Reassures users that non-English speakers will not be left behind.

3. **Four-Step Vertical Journey:**

- Step 1: Tell us your goal (~1 min)

- Step 2: Keep documents ready (~2 mins)

- Step 3: Simple verification (~3 mins)

- Step 4: Invest your first ₹100 (~1 min)

- Uses the **exact same terminology and time estimates** as the onboarding stepper to create zero cognitive rupture when transitioning from web to app.

4. **Static Illustrative Example (No Promised Growth):**

- Clearly badged: "*Example only, not a prediction*".

- Explains how ₹100 is split across index funds, government bonds, and emerging companies. Contains no hypothetical 15% return curves that mislead first-time buyers.

5. **Trust Placeholders & Native `**

▶ Details